UNIQUENESS OF THE SPLITTING OF A SIMPLICIAL SET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: two small nameable steps: the initial observation is definitional from Definition definition-split; the final step (phi xi = id for every right inverse xi of psi forces psi = phi) is a two-line argument - for each y pick a section sending psi(y) to y.

Editorial note. The label lemma-splitting-simplicial-sets names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Splitting of a simplicial set. Same unit, Definition definition-split. Uniqueness is relative to the notion of splitting from Definition definition-split; the initial observation that N(U_m) must be the nondegenerate simplices reads off that definition.

Lemma 0.1. *Let U be a simplicial set. Then U has a unique splitting with $N(U_m)$ equal to the set of nondegenerate m-simplices.*

Proof. From the definition it follows immediately, that if there is a splitting then $N(U_m)$ has to be the set of nondegenerate simplices. Let $x \in U_n$. Suppose that there are surjections $\varphi : [n] \to [k]$ and $\psi : [n] \to [l]$ and nondegenerate simplices $y \in U_k$, $z \in U_l$ such that $x = U(\varphi)(y)$ and $x = U(\psi)(z)$. Choose a right inverse $\xi : [l] \to [n]$ of ψ, i.e., $\psi \circ \xi = \mathrm{id}_{[l]}$. Then $z = U(\xi)(x)$. Hence $z = U(\xi)(x) = U(\varphi \circ \xi)(y)$. Since z is nondegenerate we conclude that $\varphi \circ \xi : [l] \to [k]$ is surjective, and hence $l \geq k$. Similarly $k \geq l$. Hence we see that $\varphi \circ \xi : [l] \to [k]$ has to be the identity map for any choice of right inverse ξ of ψ. This easily implies that $\psi = \varphi$. □

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